

# Metric Framework and First Variation of the Volume-Radius Foliation (Route $\delta$ , Plan 2, Building Block METRICFRAMEWORK)

## Abstract

We set up the metric ambient space  $(\mathcal{X}, d_{\mathcal{X}})$  of  $L^1$ -equivalence classes modulo translations, parametrize the torsion super-level sets by the volume radius, and prove the metric first-variation estimate (T) for the rescaled curve  $\rho \mapsto [F_{\rho}] \in \mathcal{X}$  in finite-perimeter generality, with no graph parametrization of  $\partial^* E_{\rho}$ . The proof has three ingredients: (i) the Sobolev–Sard theorem of Figalli–Maggi 2011 (Appendix A, Theorem A.1), which makes the normal velocity  $V_{\rho} = -t'(\rho)/|\nabla u|$  pointwise defined  $\mathcal{H}^{n-1}$ -a.e. on  $\partial^* E_{\rho}$  for a.e.  $\rho$ ; (ii) the moving-region first-variation calculus for smooth nested flows, extended to finite-perimeter levels by BV/coarea approximation, perimeter lower semicontinuity, Reshetnyak lower semicontinuity for linear-growth integrands, and the Ambrosio–Gigli–Savaré lower semicontinuity of the metric derivative; (iii) the translation-volume identity for finite-perimeter sets, which shows that the translation infimum on the right-hand side of (T) is precisely the metric-quotient cost. The centre field  $z_{\rho}$  enters as an arbitrary Borel measurable map; in the downstream chain it is instantiated as the centroid  $\bar{z}_{\rho}$  of  $E_{\rho}$ , supplied by Building Block CENTROIDBOUND. We state honestly that the pointwise-in- $\rho$  squared form is conditional on a per- $\rho$  radial trace that is open in the no-graph setting; the paper downstream uses only the integrated form, proved unconditionally in Building Block SPACETIMEIDENTITY.

## 1 Standing hypotheses

Let  $n \geq 2$  and let  $\Omega \subset \mathbb{R}^n$  be a bounded open set with  $|\Omega| = \omega_n = |B_1|$  and  $\Omega \subset B_R$ . Let  $u \in H_0^1(\Omega) \cap C^{0,1}(\bar{\Omega})$  be the torsion potential, the unique variational solution of

$$-\Delta u = 1 \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega.$$

Standard elliptic regularity (Stampacchia, see e.g. [3] §26) gives  $u \in W_{\text{loc}}^{2,p}(\Omega)$  for every  $p < \infty$ . Fix  $\rho_* \in (0, 1)$ . The volume-radius parametrization is

$$E_{\rho} := \{u > t(\rho)\}, \quad |E_{\rho}| = \omega_n \rho^n, \quad \rho \in [\rho_*, 1], \quad (1)$$

with  $\rho \mapsto t(\rho)$  absolutely continuous on  $[\rho_*, 1]$  and  $t'(\rho) \leq 0$  a.e. (Maggi’s coarea formula, [3] Thm. 13.1 and [2] Thm. 3.40). For a.e.  $\rho$ ,  $E_{\rho}$  has finite perimeter, reduced boundary  $\partial^* E_{\rho}$ , outer unit normal  $\nu_{\rho}$ , and  $P(E_{\rho}) = \mathcal{H}^{n-1}(\partial^* E_{\rho})$ . We write  $P_{\rho} := P(B_{\rho}) = n\omega_n \rho^{n-1}$ .

## 2 The metric space $\mathcal{X}$ and the rescaled curve

**Definition 2.1** ( $L^1$  modulo translations). Let  $\mathfrak{M}$  denote the set of  $\mathcal{L}^n$ -equivalence classes of characteristic functions  $\mathbf{1}_A$  with  $|A| < \infty$ , equipped with the  $L^1$  distance  $d_{L^1}(\mathbf{1}_A, \mathbf{1}_C) = |A \Delta C|$ . The group  $\mathbb{R}^n$  acts on  $\mathfrak{M}$  by translation. The metric space  $(\mathcal{X}, d_{\mathcal{X}})$  is the quotient

$$\mathcal{X} := \mathfrak{M}/\mathbb{R}^n, \quad d_{\mathcal{X}}([A], [C]) := \inf_{a \in \mathbb{R}^n} |A \Delta (C + a)|. \quad (2)$$

**Lemma 2.2** (Attainment of the translation infimum). *If  $A, C \subset \mathbb{R}^n$  have finite measure and are both contained in some ball  $B_M$ , the infimum in (2) is attained at some  $a_* \in \overline{B_{2M}}$ .*

*Proof.* For  $|a| \geq 2M$ ,  $(C + a) \cap A = \emptyset$  and  $|A\Delta(C + a)| = |A| + |C|$ , which exceeds  $|A| + |C| - |A \cap C| = |A\Delta C| + |A \cap C| \geq |A\Delta C|$ , so the infimum is unchanged if we restrict to  $a \in \overline{B_{2M}}$ . The map  $a \mapsto |A\Delta(C + a)|$  is continuous on  $\mathbb{R}^n$  (by dominated convergence), hence attains its minimum on the compact set  $\overline{B_{2M}}$ .  $\square$

**Definition 2.3** (Rescaled curve). For  $\rho \in [\rho_*, 1]$ , define the volume-rescaled level set and its class

$$F_\rho := \rho^{-1}E_\rho \subset \mathbb{R}^n, \quad [F_\rho] \in \mathcal{X}.$$

By construction  $|F_\rho| = \omega_n = |B_1|$  for all  $\rho$ . Since  $E_\rho \subset B_R$ , we have  $F_\rho \subset B_{R/\rho_*}$  uniformly in  $\rho \in [\rho_*, 1]$ , and Lemma 2.2 applies to all pairs  $([F_{\rho_1}], [F_{\rho_2}])$ .

**Definition 2.4** (Metric derivative; Ambrosio–Gigli–Savaré [1, Thm. 1.1.2]). A curve  $\gamma : I \rightarrow \mathcal{X}$  is absolutely continuous if there is  $m \in L^1(I)$  with  $d_{\mathcal{X}}(\gamma(s), \gamma(\sigma)) \leq \int_\sigma^s m$  for  $\sigma \leq s$  in  $I$ . The metric derivative

$$|\dot{\gamma}|(\rho) := \lim_{h \rightarrow 0} \frac{d_{\mathcal{X}}(\gamma(\rho + h), \gamma(\rho))}{|h|}$$

exists for a.e.  $\rho \in I$  and is the smallest admissible  $m$ .

### 3 The normal velocity $V_\rho$ and the Sobolev–Sard step

For a.e.  $\rho \in [\rho_*, 1]$ , set

$$V_\rho(x) := \frac{-t'(\rho)}{|\nabla u(x)|} \quad \text{for } x \in \partial^* E_\rho, \quad |\nabla u|(x) > 0. \quad (3)$$

**Proposition 3.1** (Sobolev–Sard for  $W_{\text{loc}}^{2,p}$ ). *Let  $p > n$ . Then*

$$\mathcal{H}^{n-1}(\partial^* E_\rho \cap \{|\nabla u| = 0\}) = 0 \quad \text{for a.e. } \rho \in [\rho_*, 1]. \quad (4)$$

*Proof.* By Stampacchia,  $u \in W_{\text{loc}}^{2,p}(\Omega)$  for every  $p < \infty$ ; fix any  $p > n$ . The Sobolev–Sard theorem of Figalli and Maggi [6, Appendix A, Thm. A.1] states that for  $u \in W_{\text{loc}}^{2,p}$  with  $p > n$ , one has  $\mathcal{H}^{n-1}(\{u = t\} \cap \{|\nabla u| = 0\}) = 0$  for a.e.  $t \in (0, \|u\|_\infty)$ . By the De Giorgi structure theorem,  $\partial^*\{u > t\} \subset \{u = t\}$  up to an  $\mathcal{H}^{n-1}$ -null set for a.e.  $t$  ([3] Thm. 15.9; [2] Thm. 3.59). Reparametrizing by  $\rho$  via (1) yields (4).  $\square$

**Remark 3.2** (Backup citations). The same statement is recorded in De Philippis–Marini–Mukoseeva 2021, Lemma 2.4, with a self-contained proof. The classical Brothers–Ziemer 1988, Lemma 2.4, gives a  $W^{2,p}$  Sard statement modulo a capacity-to- $\mathcal{H}^{n-1}$  conversion.

**Corollary 3.3** ( $V_\rho$  is pointwise defined and  $L^1$ ). *For a.e.  $\rho \in [\rho_*, 1]$ ,  $V_\rho$  is an  $\mathcal{H}^{n-1}$ -a.e. finite, strictly positive Borel function on  $\partial^* E_\rho$ , and*

$$\int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1} = \frac{d}{d\rho}|E_\rho| = n\omega_n \rho^{n-1} =: A_\rho. \quad (5)$$

*In particular  $V_\rho \in L^1(\partial^* E_\rho, \mathcal{H}^{n-1})$ .*

## 4 Centre fields

**Definition 4.1** (Borel centre field). A *centre field* for  $\{E_\rho\}$  is a Borel-measurable map  $z : [\rho_*, 1] \rightarrow \mathbb{R}^n$  with  $\sup_\rho |z_\rho| \leq R'$  for some  $R' = R'(n, \rho_*, R)$ . For each  $\rho \in [\rho_*, 1]$  and  $x \in \partial^* E_\rho$ , the associated *homothetic velocity* is

$$H_{z_\rho, \rho}(x) := \frac{(x - z_\rho) \cdot \nu_\rho(x)}{\rho}. \quad (6)$$

**Remark 4.2** (Centre identification: present block uses the centroid). This building block treats  $z_\rho$  as an abstract Borel centre field. Downstream the Plan 2 chain instantiates  $z_\rho$  as the centroid  $\bar{z}_\rho := |E_\rho|^{-1} \int_{E_\rho} x \, dx$ , supplied by Building Block [9]; that block establishes a quantitative  $H^1$ -type bound on  $\rho \mapsto \bar{z}_\rho$ .

## 5 The metric first-variation theorem (T)

**Theorem 5.1** (Metric first variation modulo translations). *Under the standing hypotheses of Section 1 and for any Borel centre field  $z = \{z_\rho\}$  with  $\sup_\rho |z_\rho| \leq R' < \infty$ , the curve  $\rho \mapsto [F_\rho]$  is absolutely continuous from  $[\rho_*, 1]$  into  $(\mathcal{X}, d_{\mathcal{X}})$ , and there is a constant  $C_{n, \rho_*, R} > 0$  such that*

$$|\dot{F}_\rho|_{\mathcal{X}} \leq C_{n, \rho_*, R} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| \, d\mathcal{H}^{n-1} \quad (7)$$

for a.e.  $\rho \in [\rho_*, 1]$ .

### 5.1 Smooth flows: the moving-region identity

Suppose temporarily that  $\Omega$  is smooth,  $u \in C^\infty(\bar{\Omega})$ , and  $|\nabla u| > 0$  on the strip  $\bigcup_{\rho \in [\rho_*, 1-\eta]} \partial E_\rho$  for some  $\eta > 0$ . The homothety  $\Psi_\rho(y) := z_\rho + \rho^{-1}(y - z_\rho)$  maps  $E_\rho$  onto  $z_\rho + \rho^{-1}(E_\rho - z_\rho)$ .

**Lemma 5.2** (Pulled-back normal velocity). *In the smooth case, the normal velocity of the rescaled-and-translated flow  $\rho \mapsto F_\rho - \rho a$ , pulled back to  $\partial E_\rho$  via  $\Psi_\rho$ , equals*

$$W_\rho^{z_\rho, a} := \rho^{-1}(V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho) \quad \text{on } \partial E_\rho. \quad (8)$$

**Proposition 5.3** (Smooth case of (T)). *In the smooth setting above,*

$$\limsup_{h \rightarrow 0} \frac{d_{\mathcal{X}}([F_{\rho+h}], [F_\rho])}{|h|} \leq \rho_*^{-n} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| \, d\mathcal{H}^{n-1}. \quad (9)$$

The proof is the first-variation identity [3, §17.3] applied to  $\tilde{F}_\rho := F_\rho - \rho a$ , followed by the homothety Jacobian (factor  $\rho^{1-n}$ ) and the translation infimum.

### 5.2 BV/coarea approximation and finite-perimeter passage

**Lemma 5.4** (Smooth approximation). *There exist  $u_k \in C^\infty(\mathbb{R}^n)$  with  $u_k \rightarrow u$  in  $W^{1,p}(\mathbb{R}^n)$  for every  $p \in (1, \infty)$  and  $u_k \rightarrow u$  pointwise a.e. The associated super-level sets  $E_t^k := \{u_k > t\}$  and reparametrizations  $|E_\rho^k| = \omega_n \rho^n$  via  $t_k(\rho)$  satisfy, for a.e.  $\rho \in [\rho_*, 1]$ ,*

$$E_\rho^k \rightarrow E_\rho \text{ in } L^1, \quad t_k(\rho) \rightarrow t(\rho), \quad P(E_\rho^k) \rightarrow P(E_\rho).$$

**Lemma 5.5** (Reshetnyak lower semicontinuity). *For every  $z \in \mathbb{R}^n$  and  $a \in \mathbb{R}^n$ , the convex linear-growth integrand  $\Phi(s, \nu) := |s - (x - z) \cdot \nu / \rho - a \cdot \nu|$  satisfies, for a.e.  $\rho \in [\rho_*, 1]$ ,*

$$\liminf_{k \rightarrow \infty} \int_{\partial^* E_\rho^k} |V_\rho^k - H_{z, \rho} - a \cdot \nu_\rho^k| \, d\mathcal{H}^{n-1} \geq \int_{\partial^* E_\rho} |V_\rho - H_{z, \rho} - a \cdot \nu_\rho| \, d\mathcal{H}^{n-1}. \quad (10)$$

**Lemma 5.6** (AGS lower semicontinuity of the metric derivative). *Let  $\gamma_k : [\rho_*, 1] \rightarrow \mathcal{X}$  be AC curves with  $\gamma_k(\rho) \rightarrow \gamma(\rho)$  in  $\mathcal{X}$  for every  $\rho$ , and  $|\dot{\gamma}_k|(\rho) \leq m_k(\rho)$  a.e. with  $m_k \rightharpoonup m$  weakly in  $L^1([\rho_*, 1])$ . Then  $\gamma$  is AC and  $|\dot{\gamma}|(\rho) \leq m(\rho)$  for a.e.  $\rho \in [\rho_*, 1]$ . ([1] Thm. 1.1.2)*

*Proof of Theorem 5.1.* By Lemma 5.4, after extracting a subsequence we may assume  $E_\rho^k \rightarrow E_\rho$  in  $L^1$  and  $P(E_\rho^k) \rightarrow P(E_\rho)$  for a.e.  $\rho \in [\rho_*, 1]$ . By Proposition 5.3 applied to the smooth approximating flow  $\{E_\rho^k\}$ ,

$$|\dot{F}_\rho^k|_{\mathcal{X}} \leq m_k(\rho) := \rho_*^{-n} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho^k} |V_\rho^k - H_{z_\rho, \rho} - a \cdot \nu_\rho^k| d\mathcal{H}^{n-1}. \quad (11)$$

The sequence  $\{m_k\}$  is uniformly bounded in  $L^1$ , so by Dunford–Pettis admits a weakly  $L^1$ -convergent subsequence  $m_k \rightharpoonup m$ . By Lemma 5.5,  $m(\rho) \geq \rho_*^{-n} \inf_a \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho|$  for a.e.  $\rho$ . By Lemma 5.6,  $\rho \mapsto [F_\rho]$  is AC and  $|\dot{F}_\rho|_{\mathcal{X}} \leq m(\rho)$  a.e. This is (7) with  $C_{n, \rho_*, R} = \rho_*^{-n}$ .  $\square$

## 6 Translations quotient kills the $a \cdot \nu_\rho$ mode

**Lemma 6.1** (Translation derivative formula). *For every bounded set  $E \subset \mathbb{R}^n$  of finite perimeter and every  $a \in \mathbb{R}^n$ ,*

$$\lim_{\varepsilon \rightarrow 0} \frac{|E\Delta(E + \varepsilon a)|}{\varepsilon} = \int_{\partial^* E} |a \cdot \nu_E| d\mathcal{H}^{n-1}. \quad (12)$$

([3] Thm. 13.8; [2] Thm. 3.84.)

## 7 Per- $\rho$ squared form: status and open issues

### 7.1 Pointwise (per- $\rho$ ) version: conditional

If one assumes the per- $\rho$  radial trace bound (statement (R) of the Wave-1 audit, in the source notation `gmt-inputs-for-metric-closure.md` (1.2)) (statement (R)) and the homothetic velocity defect bound (2.4),

$$\left( \int_{\partial^* E_\rho} |H_{z_\rho, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C_{n, \rho_*, R} D_I(t(\rho)), \quad (13)$$

then Theorem 5.1, the triangle inequality, and Cauchy–Schwarz combine to give the pointwise per- $\rho$  squared bound

$$|\dot{F}_\rho|_{\mathcal{X}}^2 \leq C_{n, \rho_*, R} (D_H(t(\rho)) + D_I(t(\rho))). \quad (14)$$

**Remark 7.1** (Honest status). The estimate (13) is conditional on the pointwise (R), which is open in the no-graph setting. Consequently (14) is *conditional* and not used downstream.

### 7.2 Integrated version: unconditional

The Plan 2 paper closes the bound only in its  $\mu$ -integrated form:

$$\int_{\rho_*}^1 |\dot{F}_\rho|_{\mathcal{X}}^2 d\mu(\rho) \leq C_{n, \rho_*, R} \delta_T(\Omega), \quad (15)$$

proved without per- $\rho$  (R) by the slicing-then-squaring of [10]. The centre choice required is the centroid  $\bar{z}_\rho$  of [9].

## 8 Summary and use downstream

**Theorem 8.1** (Statement of METRICFRAMEWORK). *Let  $n \geq 2$ ,  $0 < \rho_* < 1$ , and  $R \geq 2$ . Let  $\Omega \subset B_R$  be bounded open with  $|\Omega| = \omega_n$ , let  $u$  be its torsion potential, let  $\{E_\rho\}$  be the volume-radius foliation (1), and let  $F_\rho := \rho^{-1}E_\rho$  in  $\mathcal{X}$ . For every Borel centre field  $z = \{z_\rho\}$  with  $\sup_\rho |z_\rho| \leq R'$ :*

- *the curve  $\rho \mapsto [F_\rho] \in \mathcal{X}$  is absolutely continuous on  $[\rho_*, 1]$ ;*
- *the normal velocity  $V_\rho = -t'(\rho)/|\nabla u|$  is  $\mathcal{H}^{n-1}$ -a.e. finite and positive on  $\partial^* E_\rho$  for a.e.  $\rho$ ;*
- *there is a constant  $C_{n,\rho_*,R} > 0$  such that*

$$|\dot{F}_\rho|_{\mathcal{X}} \leq C_{n,\rho_*,R} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho,\rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1} \quad \text{for a.e. } \rho \in [\rho_*, 1].$$

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